PADHAKU
Science · Class 10 Science



CHAPTER 8

Heredity

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Questions · 8.1 Accumulation of Variation during Reproduction

Q 1 **If a trait A exists in 10% of a population of an asexually reproducing species and a trait B exists in 60% of the same population, which trait is likely to have arisen earlier?**

✔ ANSWER

Trait B is likely to have arisen earlier.

In asexual reproduction a variation spreads only as individuals with it keep dividing over many generations. A trait found in more individuals (60%) has had more time to spread than one found in only 10%.

Q 2 **How does the creation of variations in a species promote survival?**

✔ ANSWER

- Variations make individuals of a species slightly different from each other.
- If the environment changes (e.g. heat wave), some variants may be better suited to it and survive, while others die.
- These survivors reproduce and pass on the useful variation, so the **species as a whole survives**.

Example: bacteria that can withstand heat survive better in a heat wave.

🧪 Activity 8.1 · Free and attached earlobes

8.1 **Observe the ears of all the students in the class. Prepare a list of students having free or attached earlobes and calculate the percentage of each. Correlate the earlobe type of each student with that of their parents and suggest a possible rule for the inheritance of earlobe types.**

💡 SAMPLE ANSWER

Observation (sample): Most students have free earlobes (e.g. about 70%); fewer have attached earlobes (about 30%). Percentage = (number with that type ÷ total students) × 100.

- Children with attached earlobes may have parents who both have free earlobes.
- When both parents have attached earlobes, the children also have attached earlobes.

Rule (conclusion): Earlobe type is an inherited trait controlled by a gene with two forms. **Free earlobe is dominant** and **attached earlobe is recessive**: a child has attached earlobes only when it gets the recessive form from both parents.

🧪 Activity 8.2 · Confirming the 1:2:1 ratio

8.2 **In Fig. 8.3, what experiment would we do to confirm that the F2 generation did in fact**

have a 1:2:1 ratio of TT, Tt and tt trait combinations?

✔ ANSWER

Self-pollinate every F2 plant and grow its seeds (F3 progeny). Then count:

- Plants whose progeny are **all tall** were **TT**.
- Plants whose progeny are **tall and short in 3 : 1** were **Tt**.
- Plants whose progeny are **all short** were **tt**.

If about ¼ of F2 plants are TT, ½ are Tt and ¼ are tt, the ratio **1 : 2 : 1** is confirmed. (Tall F2 plants that breed true are about one-third of all tall F2 plants.)

⭐ In-text · Dominant and recessive traits (Fig. 8.4)

? **Work out which trait would be considered dominant and which one recessive in Fig. 8.4. (A violet-flowered pea plant is crossed with a white-flowered plant; F1 plants are all violet; on self-fertilisation, F2 shows violet : white = 3 : 1.)**

✔ ANSWER

Violet flower colour is dominant and **white flower colour is recessive**.

Reason: all F1 plants were violet (white was hidden), and white reappeared in one-quarter of the F2 plants (3 : 1 ratio).

❓ Questions · 8.2 Heredity

Q 1 **How do Mendel's experiments show that traits may be dominant or recessive?**

✔ ANSWER

- Mendel crossed pure tall (TT) and pure short (tt) pea plants. All F1 plants were **tall**; no medium-height plants appeared.
- So only one parental trait (tallness) was expressed in F1 though both were inherited.
- On self-pollinating F1 plants, the F2 had tall and short plants in **3 : 1** ratio; the short trait reappeared.
- The trait expressed in F1 (tall) is **dominant**; the trait hidden in F1 but seen in F2 (short) is **recessive**. A single T is enough for tallness, but a plant is short only when both copies are t (tt).

Q 2 **How do Mendel's experiments show that traits are inherited independently?**

✔ ANSWER

- Mendel crossed tall plants with round seeds and short plants with wrinkled seeds. All F1 plants were **tall with round seeds**.
- On self-pollinating F1, the F2 had the parental types (tall-round, short-wrinkled) **and new combinations**: tall-wrinkled and short-round.

- (Similarly for seed shape and colour: round-yellow 315, round-green 108, wrinkled-yellow 101, wrinkled-green 32, i.e. **9 : 3 : 3 : 1**.)
- New combinations can appear only if the two traits are inherited **separately (independently)** of each other.

Q 3 **A man with blood group A marries a woman with blood group O and their daughter has blood group O. Is this information enough to tell you which of the traits – blood group A or O – is dominant? Why or why not?**

✔ ANSWER

No, this information is not enough.

- If O were dominant, the daughter could get O from her mother and show O.
- If A were dominant, the father could be AO (carrying O). He passes O, the mother passes O, so the daughter is OO and shows O.

Both cases give a daughter with group O. We would need data from many more families/children to decide. (In fact, A is dominant over O.)

Q 4 **How is the sex of the child determined in human beings?**

✔ ANSWER

- Humans have 22 pairs of ordinary chromosomes and one pair of **sex chromosomes**. Women are **XX**; men are **XY**.
- The mother's egg always carries an **X** chromosome.
- Half the father's sperms carry **X** and half carry **Y**.
- Egg (X) + sperm with X → **XX → girl**; egg (X) + sperm with Y → **XY → boy**.

So the sex of the child is decided by the chromosome it gets from the **father**, and there is an equal chance of a boy or a girl.

📝 Exercises

1 **A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers, but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as**
(a) TTWW (b) TTww (c) TtWW (d) TtWw

✔ ANSWER

(c) TtWW

- About half the progeny are short (tt), so the tall parent must carry t: it is **Tt** (Tt × tt → ½ Tt tall, ½ tt short).

- All progeny have violet flowers, so the tall parent gives W to every child: it is **WW** (WW × ww → all Ww, violet).

2 **A study found that children with light-coloured eyes are likely to have parents with light-coloured eyes. On this basis, can we say anything about whether the light eye colour trait is dominant or recessive? Why or why not?**

✔ ANSWER

No, we cannot say.

- If light colour were recessive, light-eyed parents (both recessive pairs) would have light-eyed children.
- If light colour were dominant, light-eyed parents could also have light-eyed children.

Both cases fit the observation. To decide, we need data on crosses between light-eyed and dark-eyed parents and the ratio of eye colours in their children (and grandchildren).

3 **Outline a project which aims to find the dominant coat colour in dogs.**

💡 SAMPLE ANSWER

Project outline (sample):

1. Select pure-breeding dogs of two coat colours, e.g. black and white (whose families have shown only that colour for generations).

2. Cross a black dog with a white dog and record the coat colour of all the puppies (F1).

3. The colour shown by all F1 puppies is the **dominant** colour (e.g. if all are black, black is dominant).

4. To confirm, cross F1 dogs with each other and count the F2 puppies. The dominant and recessive colours should appear in about **3 : 1** ratio; the colour that reappears in ¼ is **recessive**.

5. Repeat with many litters and record results in a table to get a reliable ratio.

4 **How is the equal genetic contribution of male and female parents ensured in the progeny?**

✔ ANSWER

- Every body cell has **two copies of each chromosome** (a pair), one from each parent.
- When germ cells (sperm and egg) are made, each takes **only one chromosome from each pair**, so germ cells have half the number of chromosomes.
- At fertilisation, a sperm (half set from the father) fuses with an egg (half set from the mother), restoring the full number.

So the child gets **one set of chromosomes (and genes) from each parent**: an equal contribution from both, and the DNA of the species stays stable.

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**